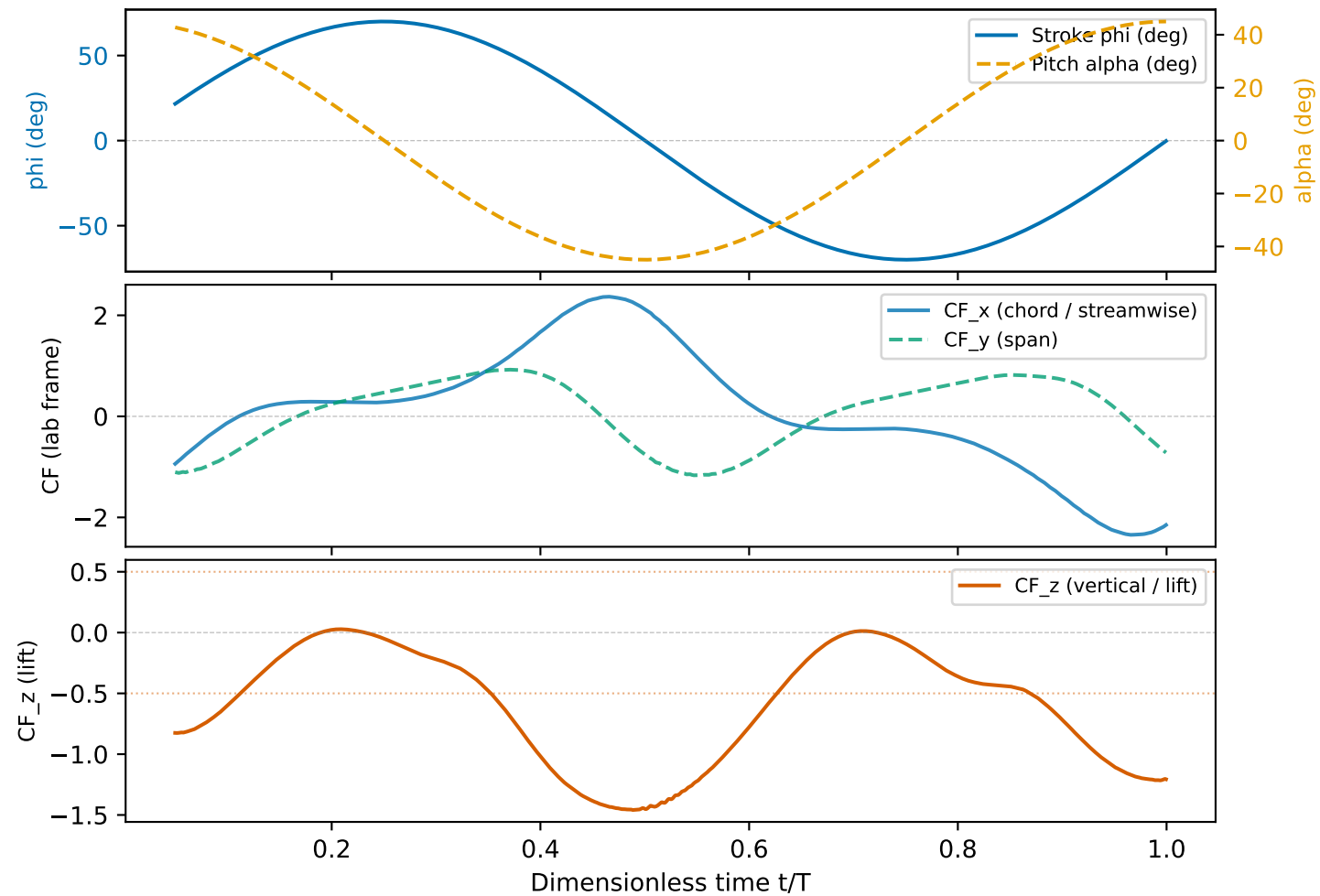


Flapping Wing Forces — coarse grid 64x32x64, $f^*=1$, $Re \sim 100$



Forces normalized by $q_{ref} \times S = 200.3$ (van Veen: $q_{ref} = 0.5 \rho u_{ref}^2$, $u_{ref} = 13.0$ at radius of gyration)